Where does Qwen3-32B hold a paper's authors?

Stages 1–9

Small samples: find a signal, try to break it

20 → 50 → 200 papers (Stage 4, Pilot50, Stage 8 cohorts)

Stages 10–13 · post-defence additions (2026-11-05) at the end

Full cohort: locate it, then intervene

9,108 papers · 149,999 name tokens (Stage 10 cohort: papers.parquet, name_tokens.parquet)

One question, four lines of argument

Each line narrows the space of explanations; small samples set it up, the full cohort settles it

dark pages = Stages 1–9 · light pages = Stages 10–11 · purple tag = comparison

Where does Qwen3-32B hold a paper's authors, and can it say them?

A · This paper?

Paper-specific, not an easy name?

- 1–9 · Stage 5, 7.2, 7.3, 8
- 10–11 · title signal, name prior, probe

B · Readout tool

What can each readout see?

- 1–9 · Stage 3, 7.4, 8, 9
- 10–11 · lens comparison, linear probe

C · Which layer

When does it appear, when is it used?

- 1–9 · Stage 4, 7.2, 7.3, 8 (L56)
- 10–11 · fine layers, patching, knockout

D · Which token

Which part of name or prompt carries it?

- 1–9 · Stage 6, 7.1
- 10–11 · name format, tail tokens, co-authors

Stages 1–9 · 20–200 papers

Pilot design

- Pilot50 / Stage 8: 50 and 200 papers (pilot50_v1.csv, stage8_papers.csv)
- Mostly first-listed author, first token
- Score: gold vs 1 or 100 false names (logit margin, win rate)
- A/B both R0, matched in pairs
- Wrong titles: 1 crossed or 20 shuffled
- Layer L56 chosen after repeated looks

Stages 10–11 · 9,108 papers

Full-cohort design

- All 9,108 papers (papers.parquet), every author, every token
- Score: rank over the whole vocabulary, log10(1/rank)
- target = A + C, control = B + D (R0 + R1), compared within citation bins
- 5 matched false titles + no title
- All layers reported, fine layers added
- Parity and Stage 6 cross-check before analysis

What changed in the design, and why

Small samples could not separate "small effect" from "no effect" in a given layer. The full cohort trades candidate-based scores for full-vocabulary ranks and pays for it with GPU time. **Note:** target/control include R1 papers; with R0 only (A vs B, as in the pilot) the output gap is +0.199 vs +0.207 (Stage 10, final L63, R0 3,423 vs 2,151; §6.13) and every layer looks the same.

Same papers, two ways of grouping

Stages 1–9 compare A vs B; Stages 10–11 compare target vs control. Counts from the 6,841-paper cohort.(Stage 1, 351_audit.json group_counts)

	recognised (target)	not recognised (control)
R0 · no true author written	A = 4,110	B = 2,186
R1 · some authors written (HR2 > 0.5)	C = 511	D = 34
Total	**target = A + C = 4,621**	**control = B + D = 2,220**

Stages 1–9

A vs B: both R0, so the contrast is purely "knows but cannot say"

Stages 10–11

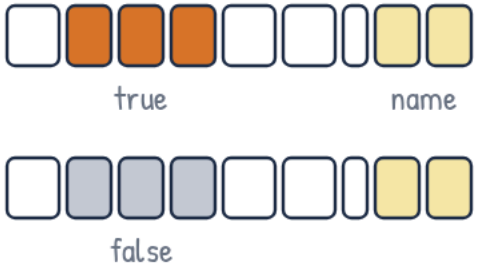
(A + C) vs (B + D): adds 511(351_audit.json) partial-recall papers, almost all on the target side

Check done: with R0 only (A vs B) the output gap is +0.199, versus +0.207 for target vs control(Stage 10, final L63, within-bin; §6.13), and every layer looks the same. The two phases compare like with like.

Reading the method figures

1 2 3 steps, in order

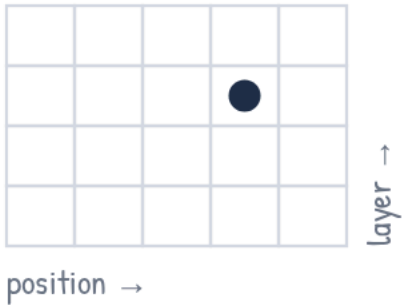
Prompt tokens



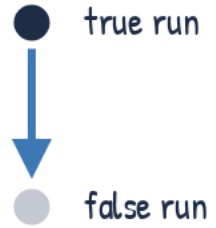
Layers



State = dot



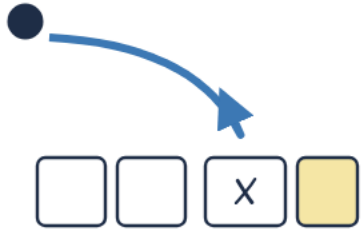
Copy a state



Block



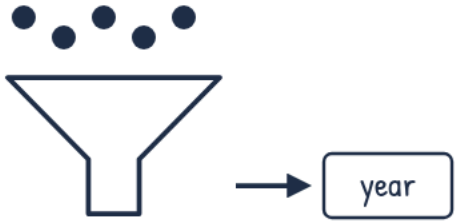
Inject



Read out



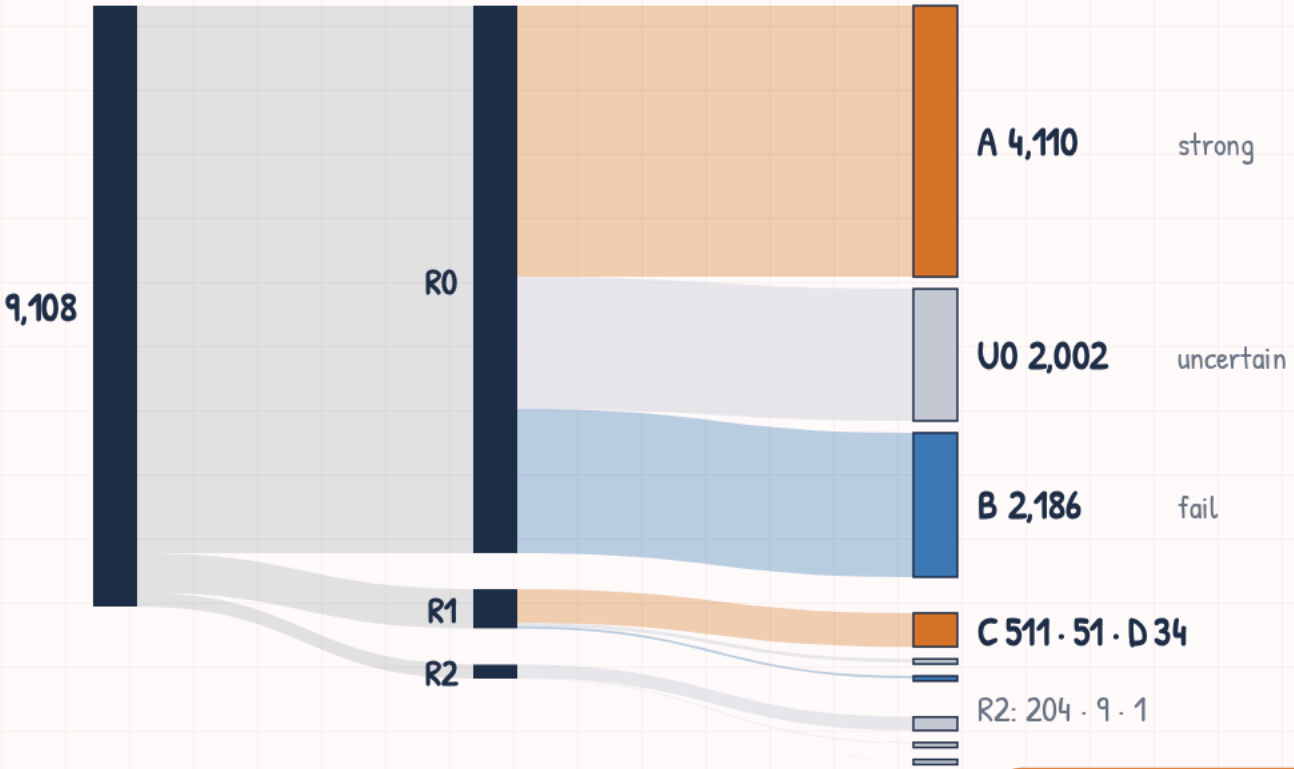
Probe



Groups and samples

example: cell C

1 papers → 2 open recall → 3 recognition → 4 cells → 5 samples



Stage 10 · M2 · M3

all papers

Stages 11–13 · X

593 target + 593 control

1,178 matched pairs

Stages 1–9

Pilot50 · Stage 8 (A, B)

target = A + C = 4,621 · control = B + D = 2,220

Counts: 351_stage10/cohort/papers.parquet (recall_level × MC outcome); 593 + 593: 361_stage11 sample (1,186 papers); 1,178 pairs: Stage 13 X1/X3 pair table

Cohort and behavioural labels

From the 15 Oct label package (9,108 papers) to a frozen 6,841-paper analysis cohort (351_audit.json total, four_group_total)

Group	Recognition / open recall	Full (351_audit.json)	Pilot50 (pilot50_v1.csv)
A	pass / R0 (no true author written)	4,110	20
B	fail / R0	2,186	20
C	pass / R1 (some authors matched)	511	7
D	fail / R1	34	3

- Pass = 4/4 option orders correct *and* mean true-option probability ≥ 0.5; fail = 0–1/4
- Pilot50: seed 35150, keeps the 20 Stage 4 papers + 30 new
- 20 A/B pairs matched on exact field and citation bin, then $|\Delta\text{year}| + 2 \cdot |\Delta\text{authors}|$

Why A vs B?

Both groups write **no** correct author. They differ only in whether the model recognises the true list. That isolates "knows but cannot say".

Caveat

In the 0–2-citation bin the strict pass rate is **25.5%** (796 papers; label_README.md l.8), above the planned ≤10% floor: "pass" is not proof of stored knowledge.

One fixed model, prompt and score

Everything later is measured under these settings

Model	Qwen/Qwen3-32B, BF16, revision 9216db57, 64 blocks, d = 5,120
Layers	L16, 24, 32, 40, 48, 56, 62 → target L63
Readouts	Public J-Lens · Vanilla (logit lens) · real final logits
Position	last input position; context limit 512
Names	encode(prompt) + encode(" " + author)
Hardware	29 blocks on GPU, 35 offloaded to CPU

> Who are the authors of the paper titled "{title}" ({year})?
> Authors:

Rank r = 1 + #tokens with a strictly larger logit

JMRR = mean over authors of mean over tokens of 1/r

Integrated = average of the 7 source layers (L63 excluded)

Guard: forced token IDs are checked, and a J-Lens/Vanilla final-logit gap > 0.1 rejects the run.

How groups and contrasts are built

labels: group lead's pipeline on the same Qwen3-32B · we use them unchanged

open answer -> R0 / R1 / R2

greedy, chat template

fuzzy name match: M, H, U

HR2 = (H+U) / (M+H+U)

R2: HR2 <= 0.5 · R0: M = 0 · R1: M > 0

a fixed rule, no human or model judge

multiple choice -> target / control

true list + 3 real teams of other papers

(same field, team size, citation bin, ~year)

4 orders · p(letter) from next token

target: 4/4 and mean p(true) >= 0.5

control: <= 1/4

5 false titles per paper

same field, own year kept

min-cost on |dyear|/5 + |dauthors|

+ |dtitle tokens| + 1.5 |dcitation bin|

no shared surname, Jaccard <= 0.6

79% strict tier (Stage 10, 35,898 of 45,540 pairs, conditions_manifest.json)

statistics

target vs control only within citation bin

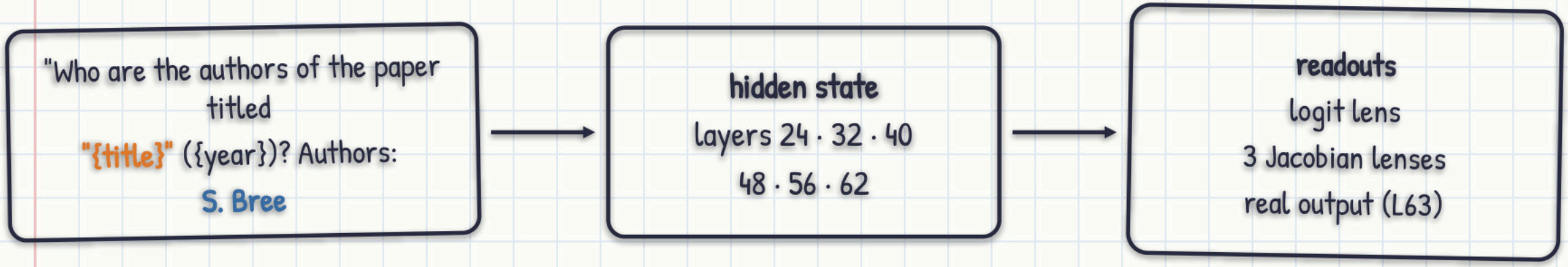
10 bins with >= 30 per group (Stage 10, §6.2)

weight $n_t n_c / (n_t + n_c)$

95% bootstrap over papers, 2,000

sensitivity: drop bins 0-4

How we read the model



Score = rank of the correct next name token,

as log10(1 / rank): 0 = rank 1, -1 = rank 10

7 title conditions: the true title, 5 matched false titles, no title

title-specific signal

= true title - mean(5 false titles)

Same name, same tokens: what is left is knowing

this paper

Lenses (model weights never change; Jacobians to L63 averaged over prompts): public = 557 WikiText-103 prompts (Neuronpedia) · self100 = 100 frozen WikiText-103 prompts · fitA = 100 other WikiText-103 prompts (Stage 9: one skipped prompt replaced -> 100 valid) (n_prompts: 351_stage10/config/lenses.json; Stage 9 final_status.json)

Can we trust the numbers?

fast = slow

One forward pass vs one per prefix: max gap 0.37 logit (bf16 rounding); ranks 1-41 identical (Stage 10, 12 contexts, parity.json)

matches Stage 6

972 tokens: same token ids, median log-prob gap ~1e-5 (Stage 10, crosscheck_stage6.csv)

negative control

Punctuation tokens show ~0 title-specific signal at every layer (Stage 10, max 0.014, 9,108 papers, §6.5)

Caveat: the parity tolerance was raised 0.25 -> 0.5 after seeing these bf16 gaps. (parity.json tolerance_amendment)

Line A

Is it about this paper?



small samples asked

Does the true title beat a crossed or shuffled title? Does A gain more than B?

→

the full cohort answered

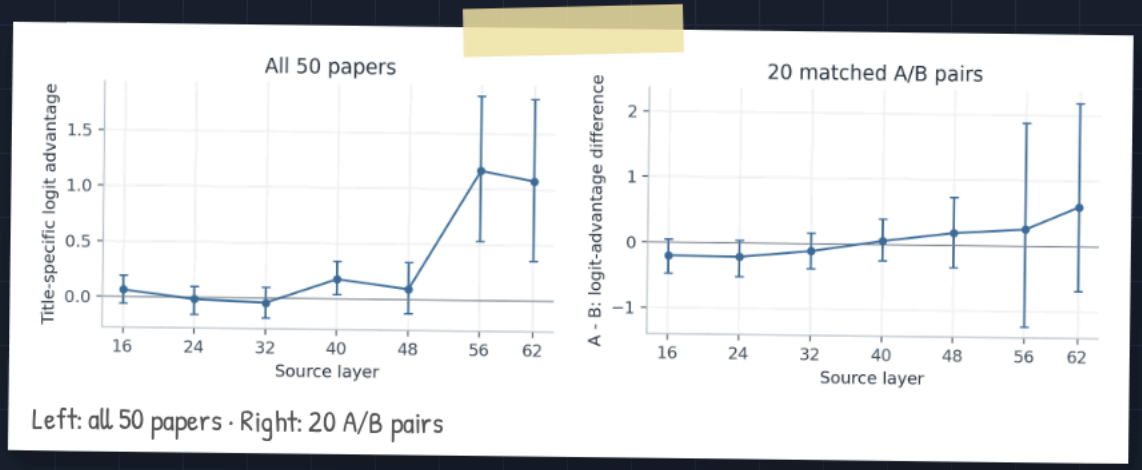
Title-specific signal for every token; target beats control only late; name prior split out; probe controls

Crossed title–author control

Is the gold name high because of the title, or because the name is easy?

Paper P (author G) and source paper Q (author F, same field & bin).
M = logit(G) − logit(F) · T = M(P title) − M(Q title)
A fixed preference for G cancels out.

- All 50 papers: L56 +1.17 [0.53, 1.84], L62 +1.08 [0.36, 1.82] (Stage 5, J-Lens, 50 papers, layer_summary.csv)
- Matched A–B, 7-layer integrated: +0.08 [−0.43, +0.65] (20 pairs, matched_ab_summary.csv)

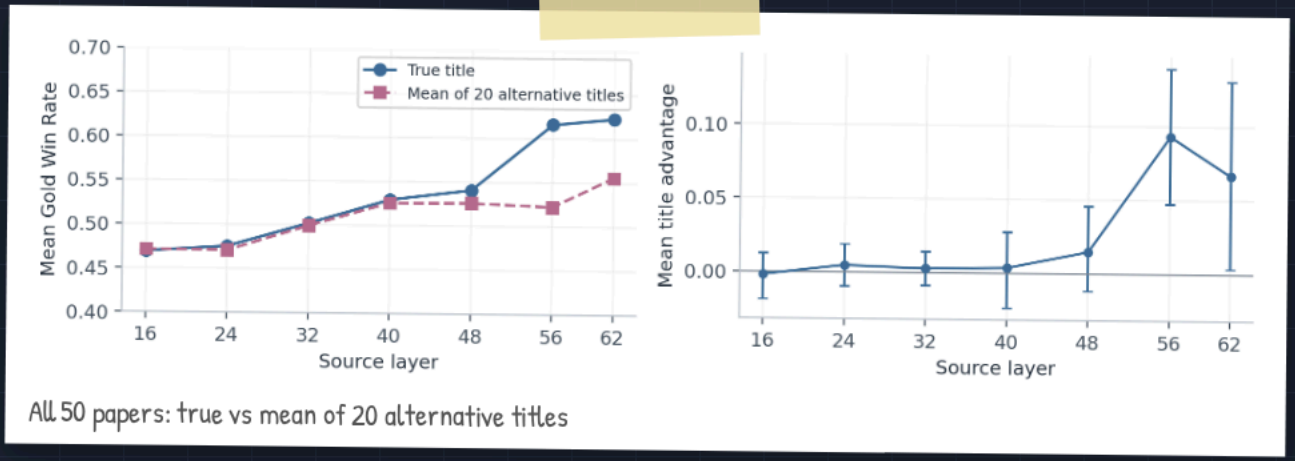


Left: all 50 papers · Right: 20 A/B pairs

The title changes which name wins in late layers. Whether A gains more than B: **not established**.

Shuffled titles: the true title helps

Same 100 candidates, 20 metadata-matched wrong titles per paper, 1,000 contexts; only the title changes



All 50 papers: true vs mean of 20 alternative titles

- Pooled L56: 0.616 vs 0.523 → +9.3 pp [4.8, 14.0], 35/50 papers positive (J-Lens L56, 50 papers; paper_null_distribution.csv; CI recomputed 10-30)
- Leave-one-out: +8.6 to +10.0 pp (stage7_l56_loo.csv)
- Vanilla L56 +10.6 pp, final output +7.2 pp (paper_null_distribution.csv)
- Matched A–B integrated: +0.018 [−0.021, +0.063] (20 pairs, integrated_ab_bootstrap.csv)

Two different questions: "does the title help?" **yes**; "does it help A more than B?" **not shown**.

Does the title effect survive the obvious objections?

Source-author overlap

199/1,000 wrong titles come from a paper whose author is in the false pool. Excluding them: +0.026 → +0.028.(quality.json; paper_null_distribution.csv)

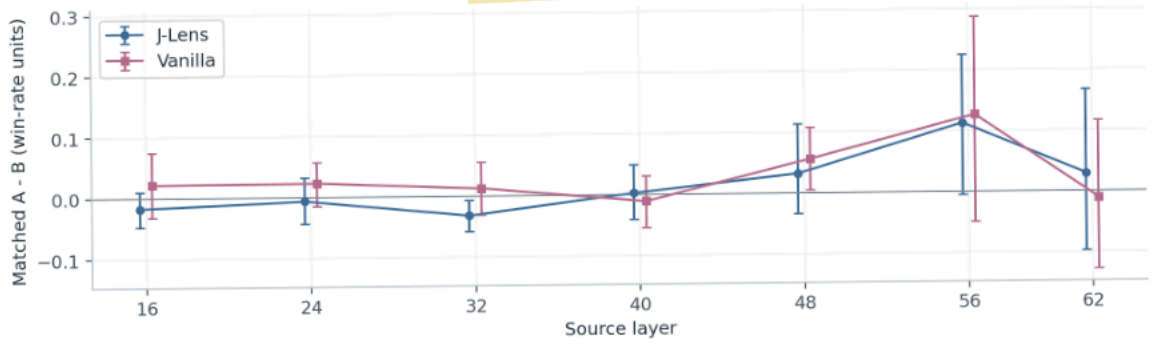
Candidate weighting

Author-, unique-token- and non-collision weights: pooled +0.026 to +0.032; all six A–B CIs include 0.

(stage7_weighting_title_advantage.csv)

Low citations

0–2 citations (n = 6): L56 +0.22, 5/6 positive, but 0/6 beyond its own 20-title tail (≤ 0.05).(floor_L56_low_six.csv)



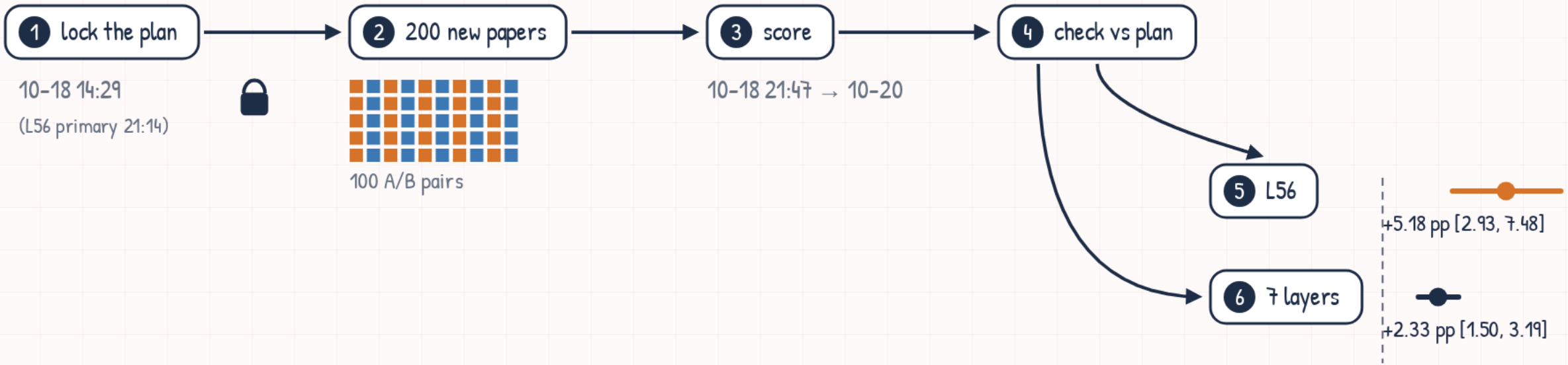
Title-adjusted A–B win rate by layer

Layer heterogeneity

L56 +0.11 but L32 **−0.03** (CI below 0).(20 pairs, matched_ab_bootstrap.csv) Reporting only the positive layers would mislead.

Independent replication (Stage 8)

plan first, data second

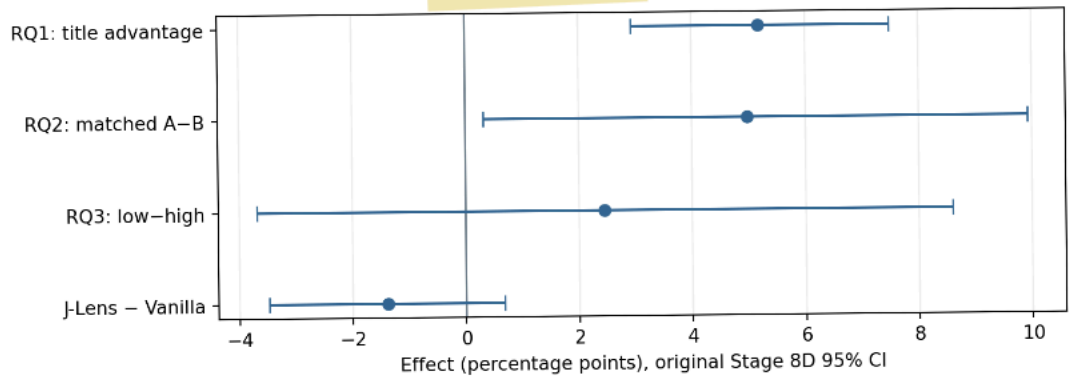


RQ1 = win rate(true title) − mean win rate(20 false titles)

first author vs 100 false names · J-Lens · paper bootstrap

+5.18 pp: Stage 8D RQ1, J-Lens L56, 200 papers, stage8D/summary/primary_questions.csv · +2.33 pp: J-Lens 7-layer, 200 papers, recompute_2026-10-30/s8_endpoints.csv

On 200 new papers the title effect replicates



Stage 8D L56 estimates (PROTOCOL.md), 95% CI (pp)(primary_questions.csv)

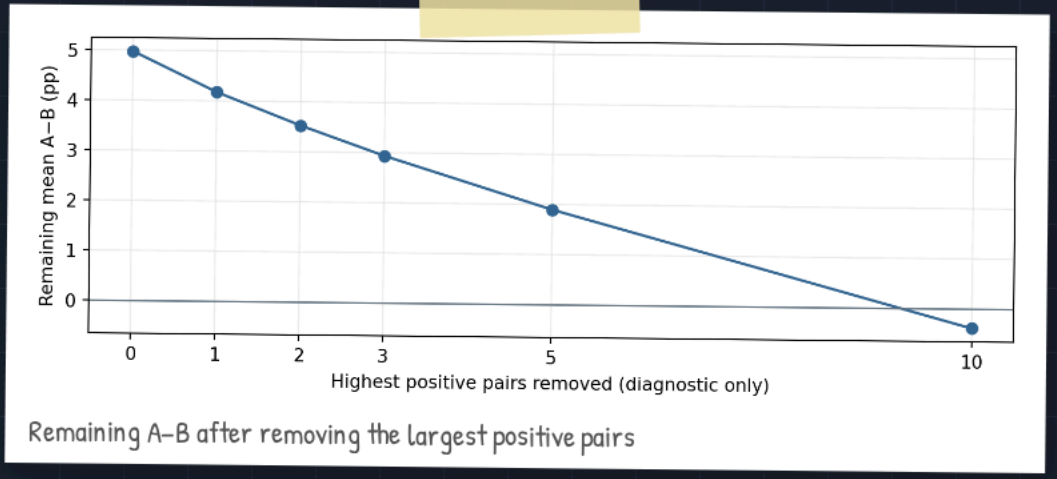
(Stage 8D, J-Lens L56; primary_questions.csv)	Estimate	95% CI
RQ1 title, L56(200 papers)	+5.18 pp	[+2.93, +7.48]
RQ2 A–B, L56(100 pairs)	+4.98 pp	[+0.32, +9.91]
RQ2 A–B, 7-layer (frozen primary)(100 pairs; s8_endpoints.csv)	+2.41 pp	[+0.91, +4.00]
RQ3 low–high(25 + 25 pairs)	+2.45 pp	[−3.69, +8.60]
J-Lens–Vanilla(jlens_minus_vanilla.csv)	−1.38 pp	[−3.48, +0.68]

Citation bands: low +6.8 [2.6, 11.6], mid +4.8, high +4.4 [−1.1, 10.0]

(50/100/50 papers; layer_and_band_summary.csv)

RQ1 is the first **independent** confirmation: correct titles help the gold author on new papers. J-Lens still adds nothing over Vanilla.

But the A–B gap rests on a few pairs



Remaining A–B after removing the largest positive pairs

- 100 pairs: 55 positive, 45 negative; median +4.4 pp (J-Lens L56; matched_pair_differences.csv)
- One pair (S8P094) ≈ +84 pp
- Remove top 5 → +1.9 pp; top 10 → −0.4 pp (recomputed from same file)

8E external audit (post-hoc)

- Unique-token weights: RQ1 +4.6, RQ2 +5.4 pp (s8_8E_weighting_by_paper.csv)
- Wrong-title length gap: +0.12 pp per token (200 papers; s8_unsourced_intervals.csv) (association only)
- Original scripts missing; recomputed from raw logits (30 Oct), same estimates (351_stage8/recompute_2026-10-30/): post-hoc only

Small samples can show a positive A–B mean that a handful of papers carry. This is the main reason Stage 10 moved to the full cohort.

Stage 8 frozen endpoint, re-scored on CPU

Addendum 13 C-2 · same 200 papers / 100 pairs · Stage 10 raw scores, all authors, all name tokens · public J-lens rank metric, NOT the W statistic

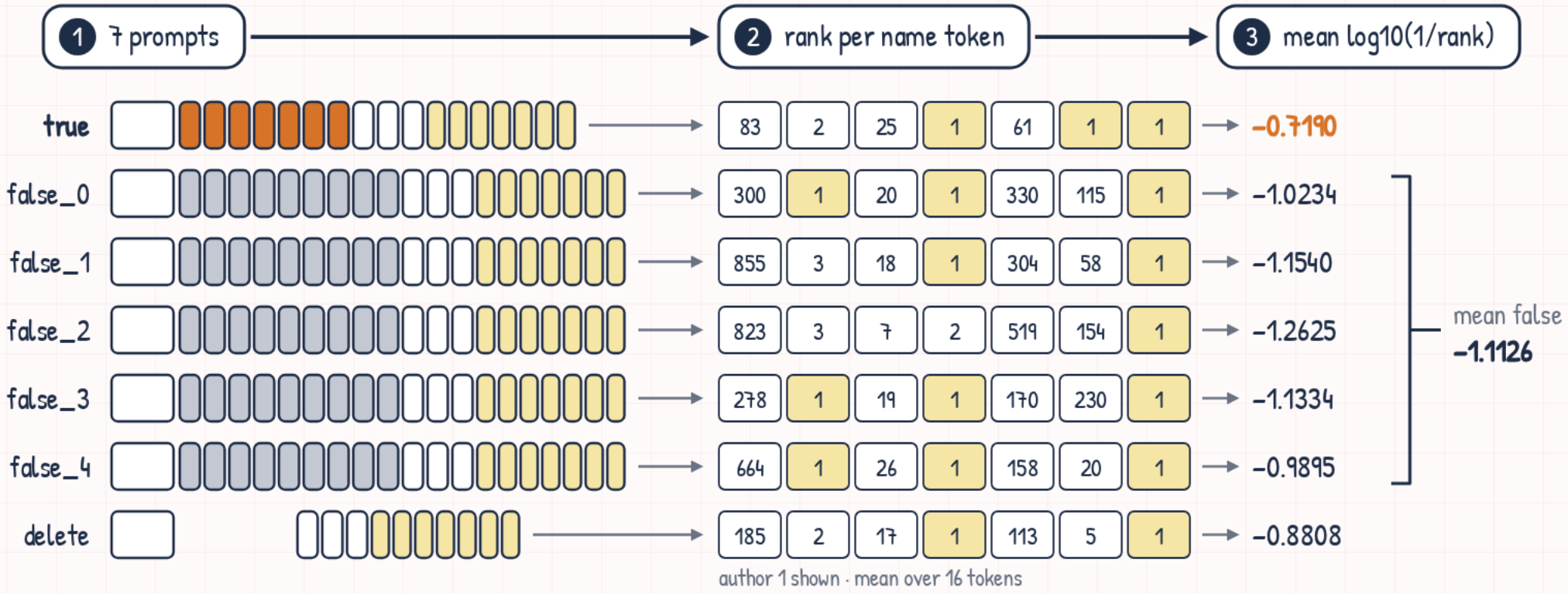
- **Multi-layer endpoint:** RQ1 +0.130 [0.109, 0.153], RQ2 +0.054 [0.008, 0.102] → both hold (public J-lens 6-layer mean; c13_C2.md aa_all)
- **L56 alone:** RQ2 +0.064 [−0.032, +0.163], not significant (same, L56)
- **What it is not:** a CPU proxy with Stage 10's 5 false titles and the rank metric; the GPU re-run of the original W was not done (user decision 2026-11-06)

Reading

The frozen multi-layer conclusion survives an all-author, all-token re-scoring; the single-layer L56 A − B gap does not.

Scoring: rank → log → spec

Coifman, Bonanno, Rafaeli (2007) · output layer



4 contrast

spec = mean nlr(true) − mean over 5 false titles

= −0.7190 − (−1.1126) = +0.3935

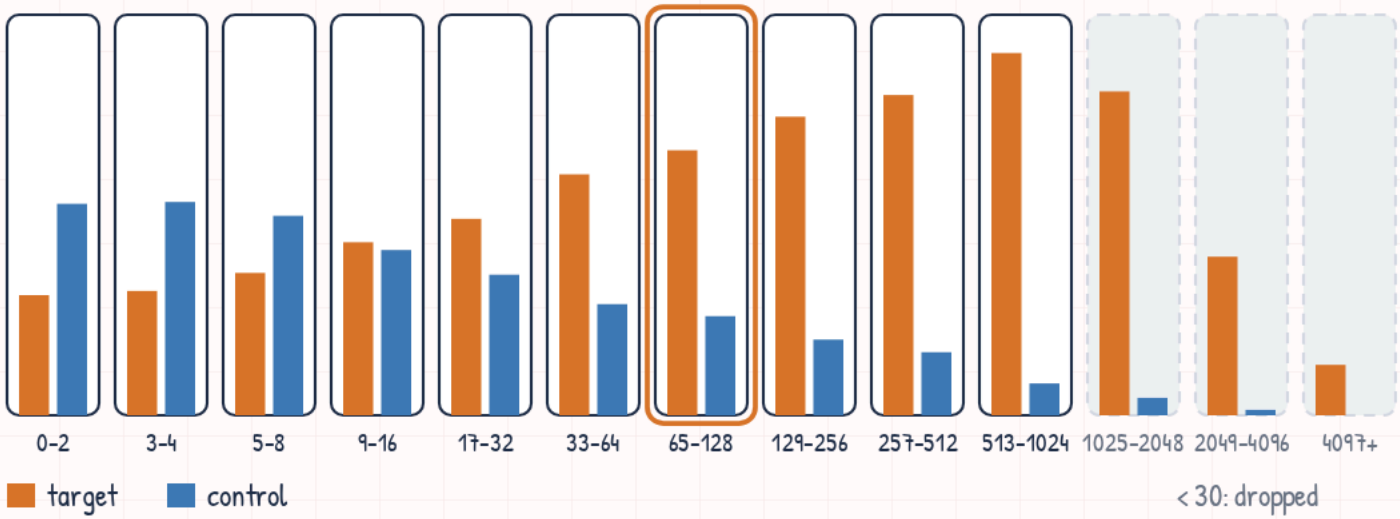
total = true − delete = +0.1618 · topic = mean false − delete = −0.2318

Ranks: 351_stage10/raw/[cond_*/]final/L63/shard_011.npz rows 25025–40 (author 1 shown) · nlr, spec, total, topic: summary_title/paper_title_contrasts.parquet (final, L63)

Statistical conventions

example paper: bin 65–128

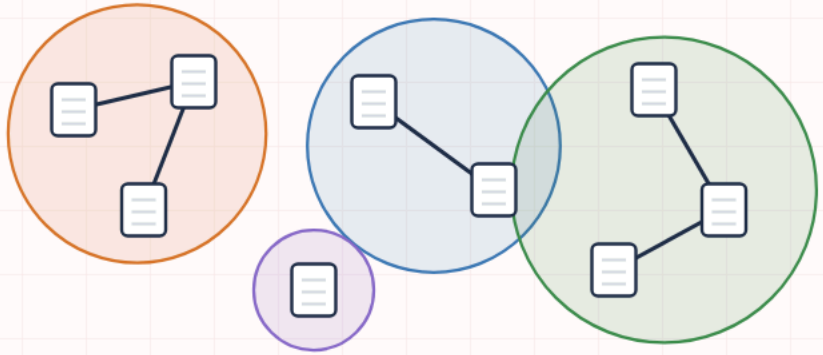
1 citation bins → 2 target − control → 3 weighted mean



< 30: dropped

$$\Delta = \Sigma\, w_b \cdot (\text{mean}_t - \text{mean}_c),\ w_b \propto n_t \cdot n_c / (n_t + n_c)$$

4 teams → 5 resample teams



shared author name = one team

7,247 teams

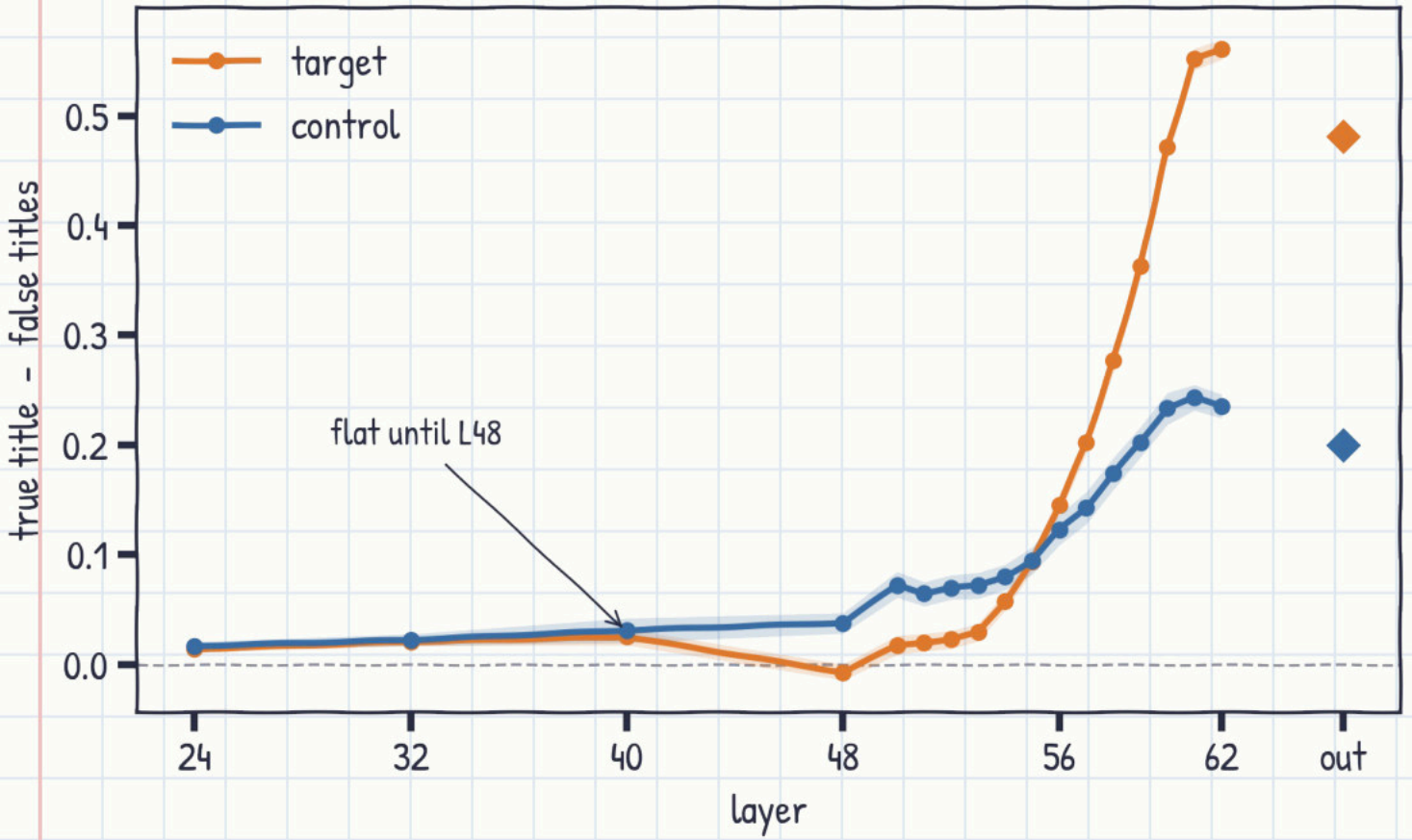


one team, one fold · 2,000 draws

Stage 10

Knowing the paper shows up only at the end

title-specific signal · public Jacobian lens (lines) and model output (diamonds)



L24–L48: ~0 for both groups (Stage 10, all readouts max 0.04, §6.3)

rises at **L56**

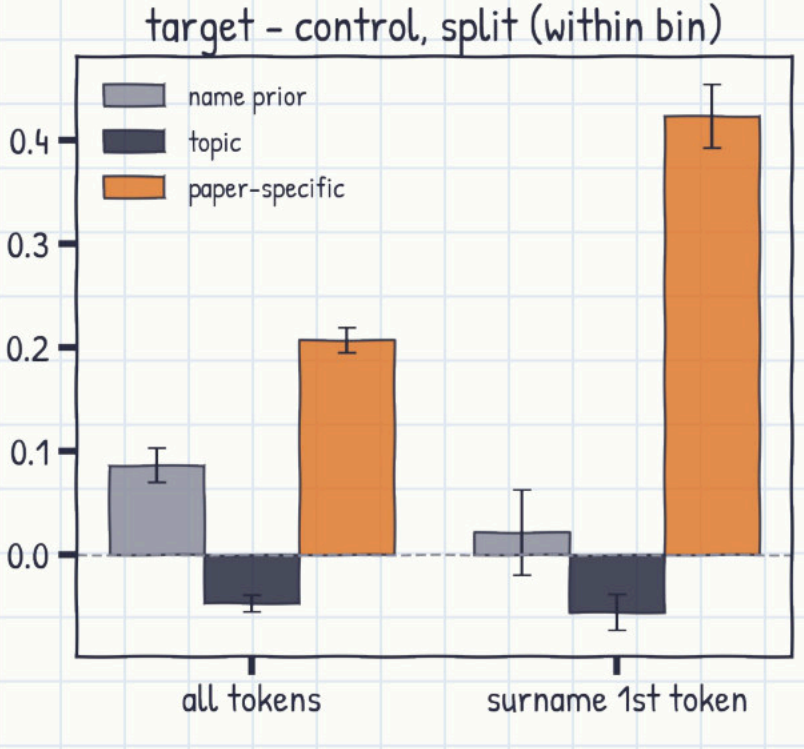
L62: **target +0.56** vs **control +0.24**
(Stage 10, public lens L62, 4,621 + 2,220 papers, §6.3 fig.)

Stage 11

How much is just the name prior?

model output · no title vs false title vs true title · all 9,108 papers





no title: rank ~12 (Stage 12 A, 9,108 papers, §6.12)

target - control: +0.09 prior, +0.21 paper (Stage 12 A, 4,621 + 2,220, within bin, §6.12)

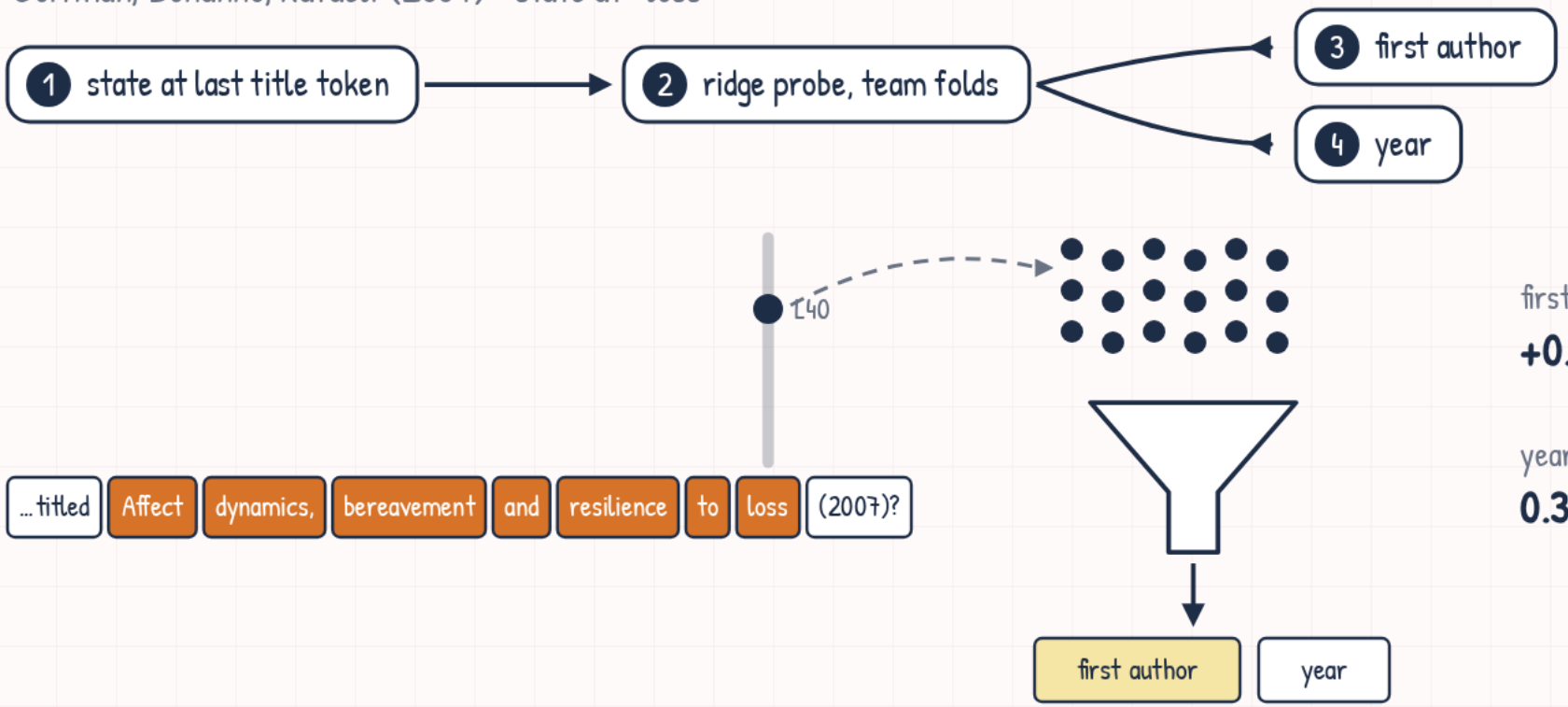
surname 1st: prior ~0, paper +0.42 (same, 1st surname token)

prior vs recognition rho 0.09 (Stage 12 B, 8,894 R0/R1 papers)

Probe

⚠ E4b scores only the first name token (mostly an initial) · all tokens by role: see the addendum-9 slide (surname first token L40 +0.071(1,395 papers; e4b_alltok.md))

Coifman, Bonanno, Rafaeli (2007) · state at "loss"



first author, L40

+0.061

year R^2, L16 → L52

0.38 → 0.56



colour = author team; one team, one fold

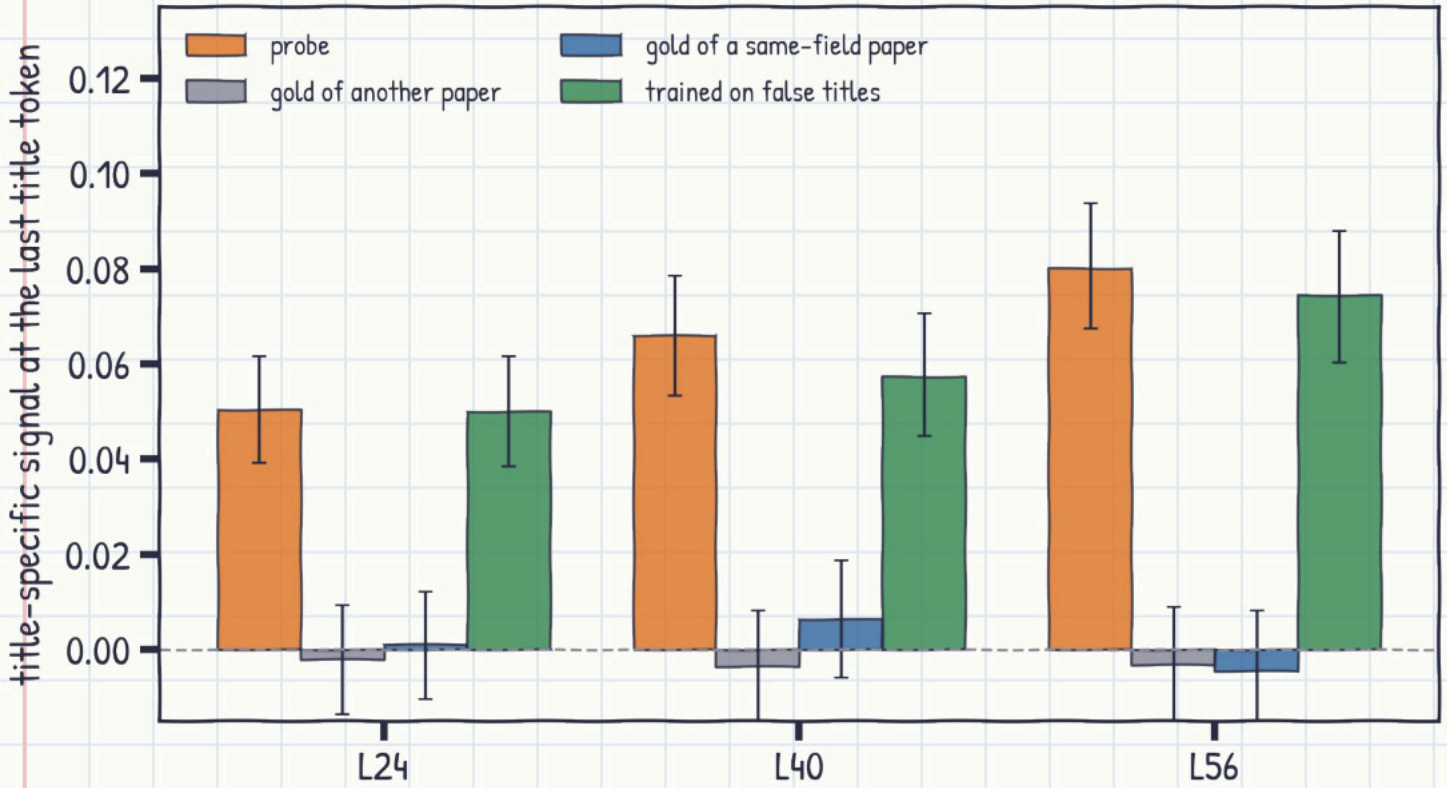
spec = probe(true title) − probe(false title)

probe score = log10(1/rank) of the true first-author token

+0.061: Stage 13 E4b re-check, probe scores ONLY THE FIRST NAME TOKEN, all papers (1,863), team folds, title reuse dropped; e4b_trainstd_compare.md
year R^2 0.38 → 0.56: Stage 13 M2, 9,108 papers, team CI; summary_cpu_teamci/M2_teamci.csv

The probe signal is paper-specific

last title token · 5-fold ridge probe · re-checked: author-team folds, title reuse removed



another paper's authors: **0**

same-field paper: **0** (Stage 12 X1 + Stage 13 re-check, §6.12)

so it is **this paper**, already at L24

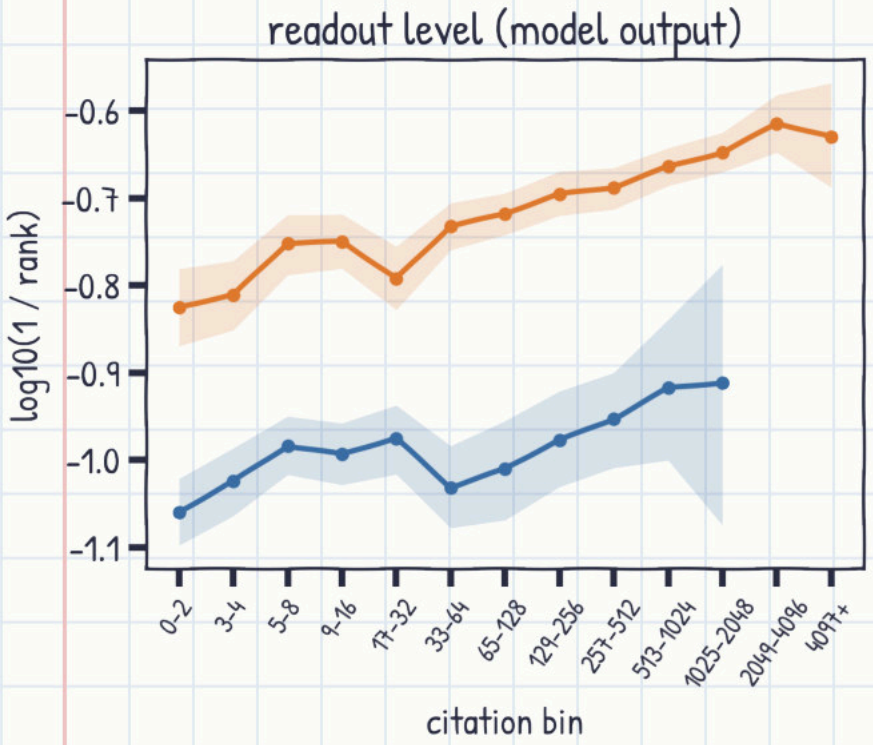
lenses cannot read it

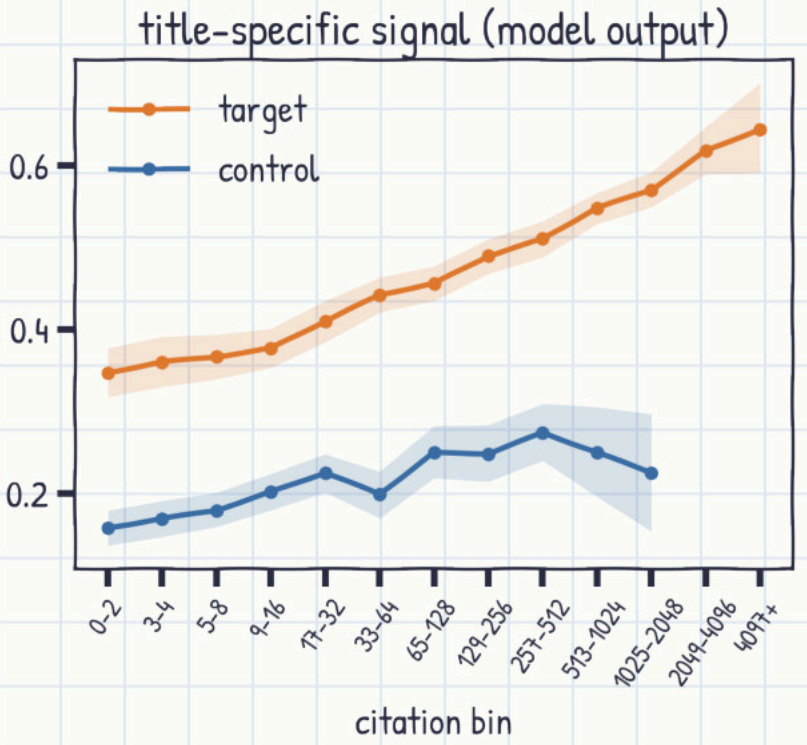
Probe scores only the first name token (mostly an initial); all-token, by-role version (addendum 9, done): surname first token L40 +0.071 [0.058, 0.085] (1,395 papers; e4b_alltok.md).

Stage 10

More citations, more paper knowledge

model output · bins with < 10 papers omitted





Target: +0.35 -> +0.64 (Stage 10, final, 200 / 84 papers, §6.8)

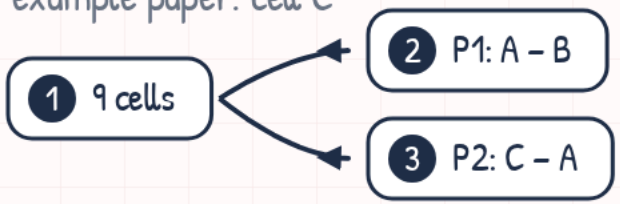
Spearman (title-specific)

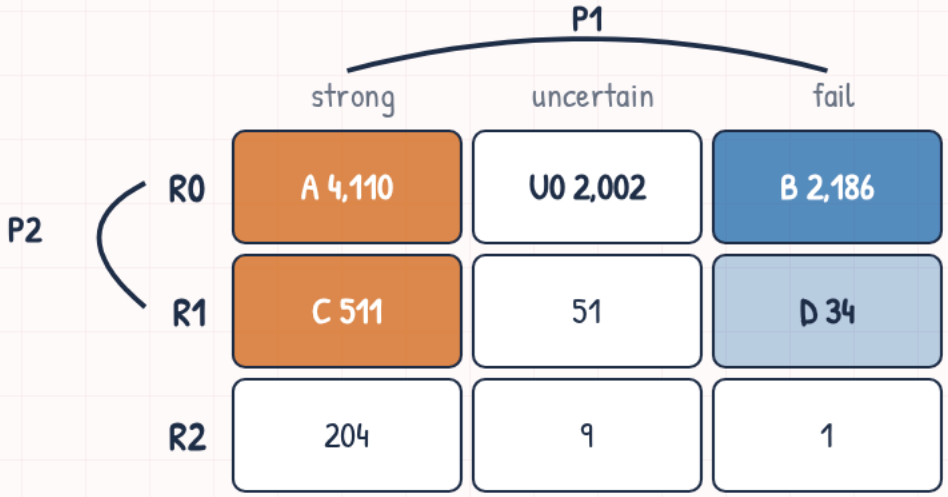
0.45 all · 0.33 target · 0.19 control (Stage 10 X2, 9,108 papers, §6.8)

Target > control in every bin

Group comparisons

example paper: cell C







P1 +0.200 [0.188, 0.213]

P2 co-authors +0.061

first author +0.031 (ns)



contrast = Σ w_b (mean_cell1 – mean_cell2)

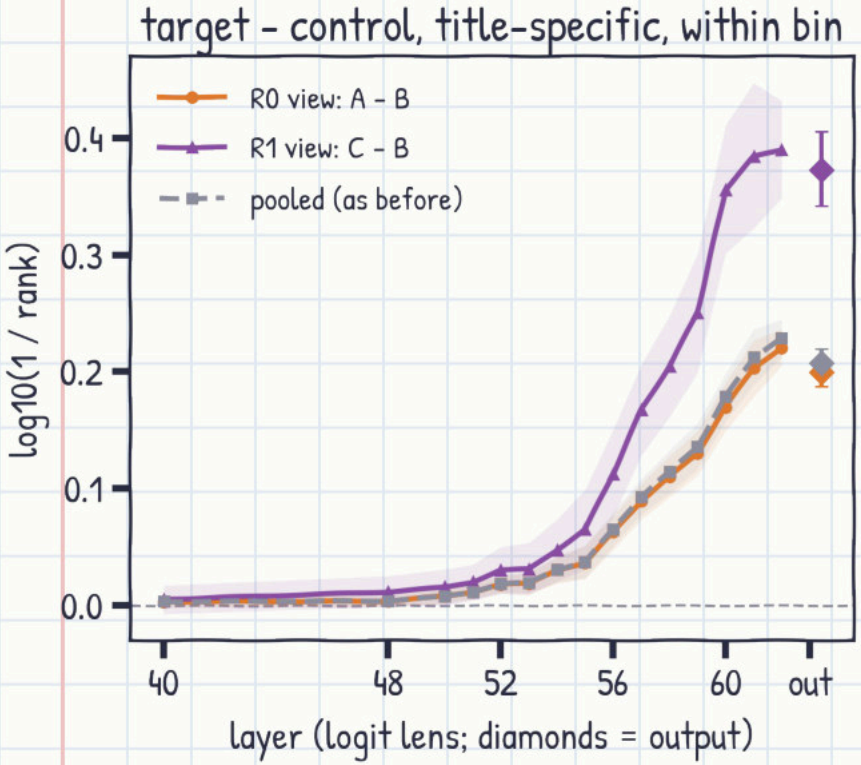
title-specific signal, unwritten authors, output layer; team bootstrap

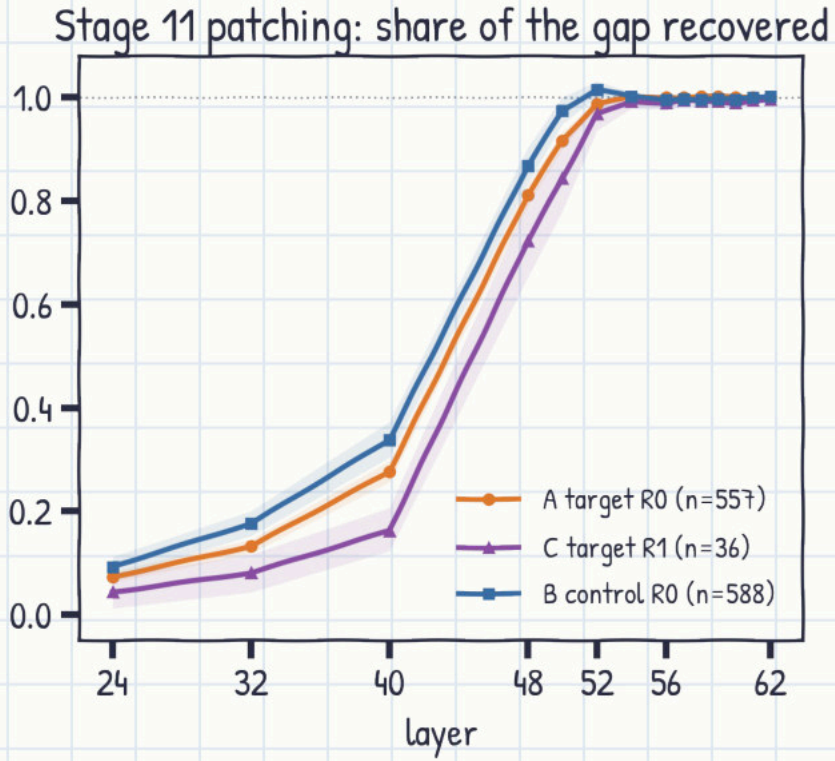
P1 +0.200 [0.188, 0.213]: A – B, 3,417 / 2,148 papers; P2 co-authors +0.061, first author +0.031: C – A; dose 0.17 → 0.46: R0 by orders right; 391_groups/primary.csv, stage10_curves.csv

Stage 10–11

Is it just the R1 papers? No.

R0 = no true author written · R1 = a few written · A/B/C/D = target/control x R0/R1





R0 only (A – B):

output +0.199 vs +0.207, same shape (Stage 10, A 3,423 / B 2,151, §6.13)

R1 (C – B): +0.37, also late (C 391 / B 885)

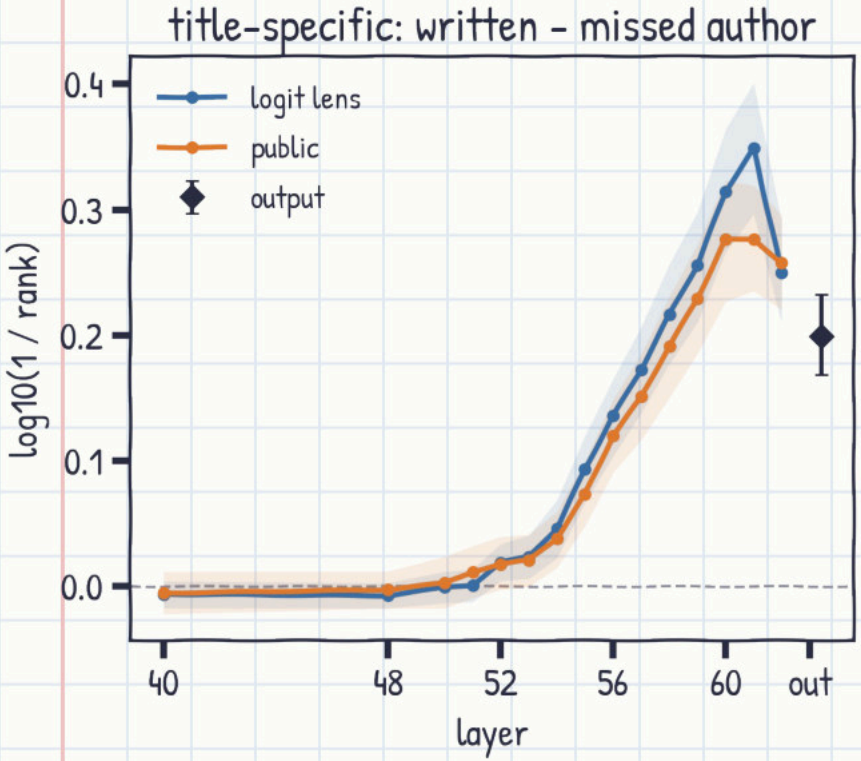
C – A within target: +0.11, from L56 (C 485 / A 3,204)

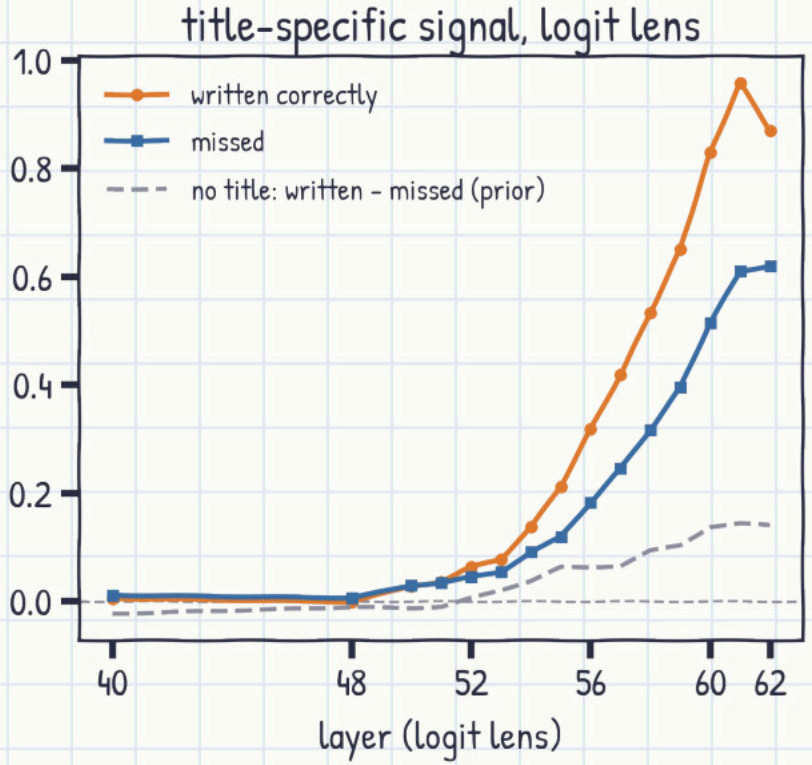
patching: R1 a bit later, done by L52 (Stage 11 E1, 36 R1 targets)

Stage 10–11

Authors it wrote vs authors it missed

inside 495 R1 papers · paired within paper · 95% CI (Stage 10, r0_r1_analysis.md §4)





written − missed, output:
+0.20 (Stage 10, final L63, 495 papers)

~0 up to L52, grows from **L54**

no title (prior): +0.10 (same, no title)

same without first authors

Stages 1–9 · 20–200 papers

What we saw

- True title helps: pilot L56 +9.3 pp (7.3, 50 papers); 200 new papers +5.2 pp (8D RQ1, 200 papers)
- A–B uncertain in the pilot; +5.0 pp on 200 papers but gone after dropping 10 pairs (8D RQ2, 100 pairs)
- Low – high citation: inconclusive

Stages 10–11 · 9,108 papers

What scale showed

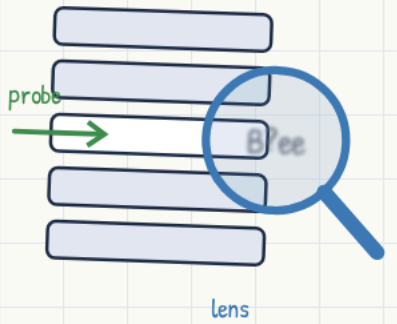
- Title-specific signal clear in every group, ≈0.44 at L62 (L62, §6.9)
- target – control ≈0 at L24–L40, slightly negative at L48 (−0.04 to −0.05) (§6.4), +0.21 [0.20, 0.22] at the output (L63, §6.4)
- Name prior is only ~1/3 of the gap; ~0 on surnames (§6.12)
- Signal rises steadily with citations (ρ = 0.451) (§6.8)
- Same groups as the pilot (R0 only, A – B): +0.20 [0.19, 0.21] (R0, §6.13), same late shape
- R1 papers (C) know more (+0.11 over A) (§6.13) but are just as late

Line A: from "the title helps" to "this paper, late, graded by fame"

The pilot was right that the title matters. It could not see that the target advantage is real but confined to the last layers, so the 7-layer average and the L56 endpoint both diluted it. Restricting the full cohort to the pilot's A/B groups (R0 only) changes nothing: the R1 papers do not drive the result.

Line B

What can each readout see?



small samples asked

Is J-Lens better than Vanilla? Is the fitted lens stable?

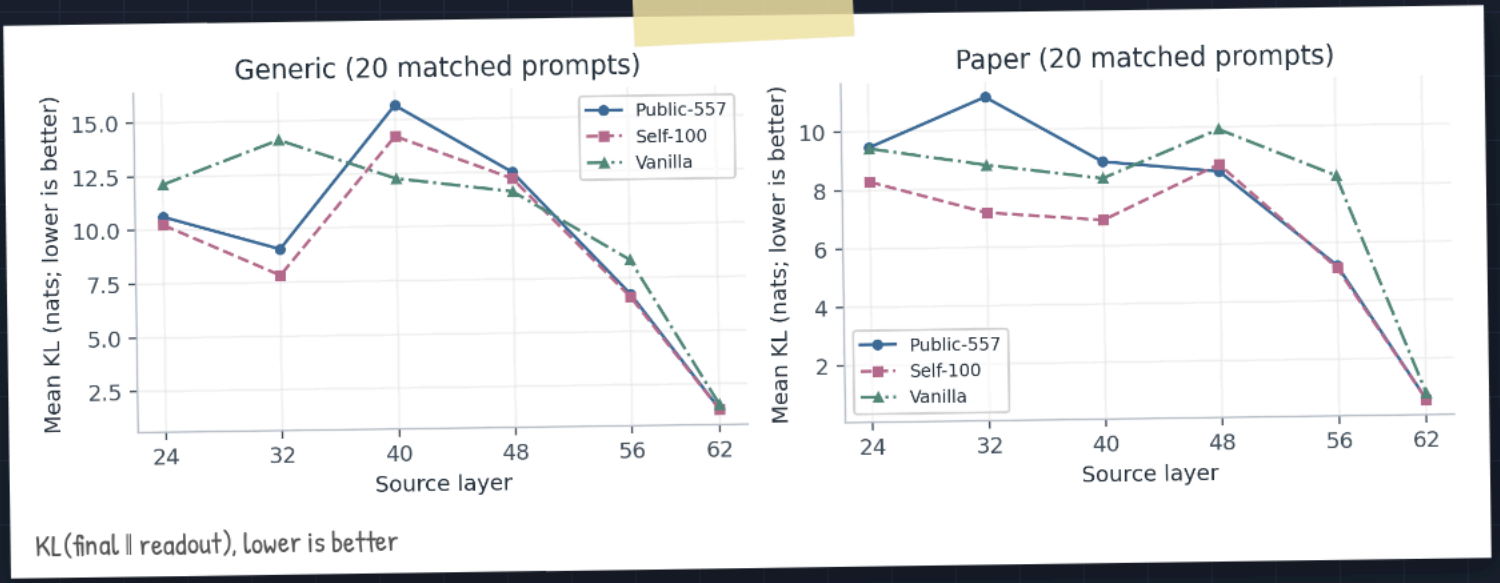
→

the full cohort answered

Lenses read name shape earlier, not paper identity; a probe at the title token reads paper-specific information

Two lenses: Public-557 vs Self-100

Can a middle layer approximate the final next-token distribution? (20 generic + 20 paper prompts)

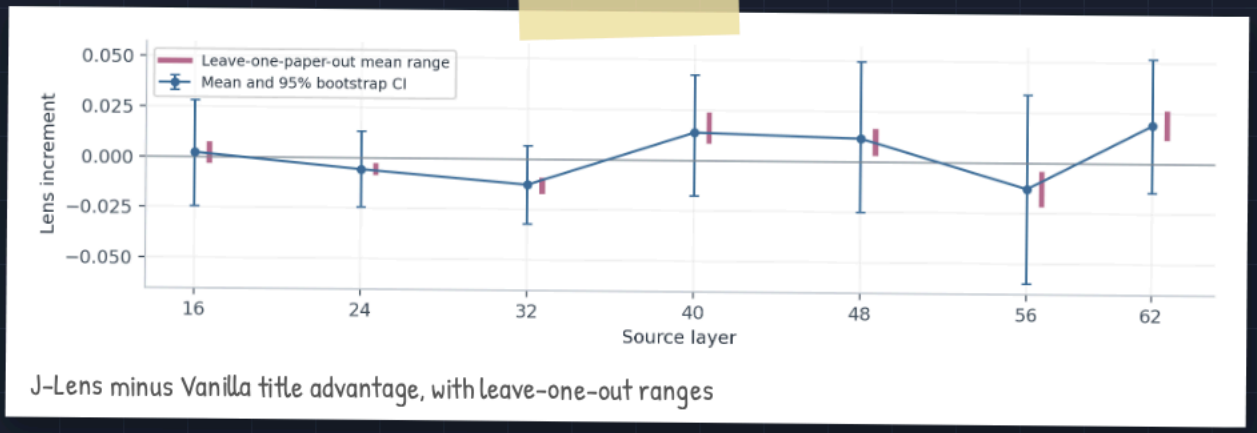


- Both checkpoints numerically valid (63/63 and 6/6 finite)(matrices.csv; frobenius_norms.csv)
- Self-100 has lower KL on 6/6 generic and 5/6 paper layers (paper L32: 11.1 → 7.2) (20 + 20 prompts; metrics.csv)
- But at paper L62, Self-100 KL beats Vanilla (0.58 vs 0.81) while Top-10 overlap is worse (0.36 vs 0.72)(metrics.csv)
- Author pilot: both lenses 14/20 full-name wins at L48(20 papers, layer_summary.csv)

Lesson: a better distribution match does not mean better author ranking. Fidelity cannot stand in for the task.

Cross-stage audit: does J-Lens add value?

Read-only joins of Stages 5–7.3; 20,000 bootstrap draws



J-Lens minus Vanilla title advantage, with leave-one-out ranges

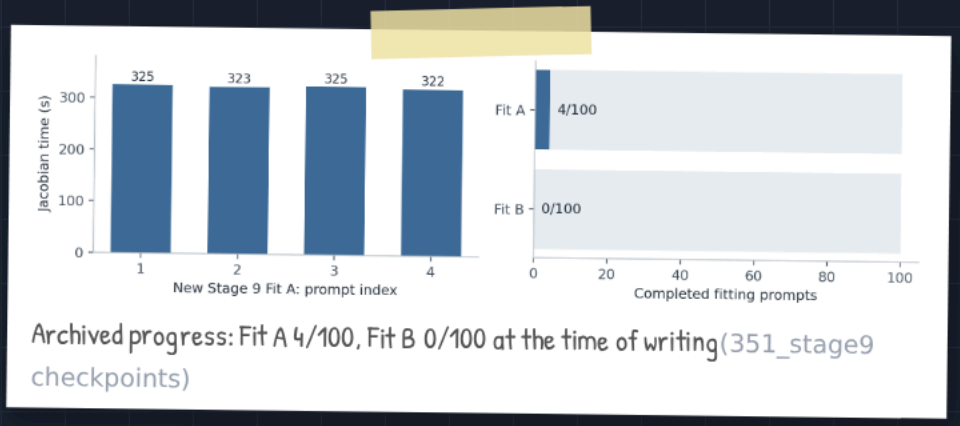
Stage 5 vs 7.3 at L56	r = 0.45, ρ = 0.47; signs agree 32/50 (50 papers, cross_stage_correlations.csv)
J-Lens − Vanilla	+0.002 [−0.008, +0.012]; median −0.002; 24/50 positive (jlens_increment_bootstrap.csv)
A−B of that increment	−0.014 [−0.033, +0.004] (20 pairs)

Dropping one paper can flip the sign of the J-Lens increment. The pilot shows **no added title-specific value** from J-Lens over Vanilla. Everything so far is exploratory: L56 was inspected repeatedly.

Stage 9: is the lens stable across fitting data?

New FullFit protocol, different from the historical 50+50 split

	Historical split-half	Stage 9 FullFit
Fit data	Self-100 split 50 + 50	two disjoint 100-prompt sets
Holdout	20 generic + 20 paper	independent 100 WikiText
Layers	6	7 (adds L16)
Status	complete	only Fit A completed; Fit B and holdout not run



Archived progress: Fit A 4/100, Fit B 0/100 at the time of writing (351_stage9 checkpoints)

- WikiText-103 train, 100–1,200 chars, deduplicated by hash, seed 3519001
- ~324 s Jacobian time per prompt, ~62 GiB CPU RAM (full_A.log; benchmark log); per-prompt checkpoints

Only Fit A was finished, and Stage 10 used it as "fitA". Without Fit B and the holdout, fitting stability is **untested**.

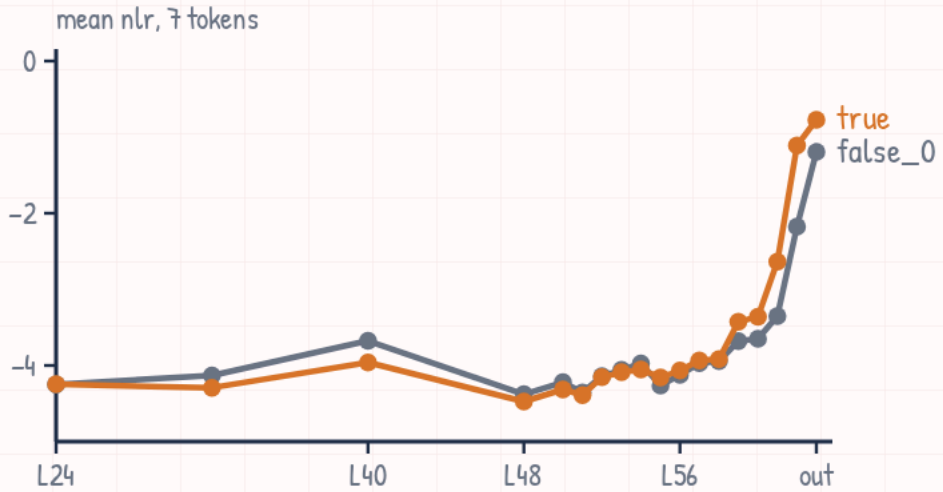
Layer-by-layer readout

Coifman, Bonanno, Rafaeli (2007) · logit lens

1 prompt → 2 state at name → 3 lens → vocabulary → 4 rank of true token → 5 mean

nlr = log10(1 / rank)

rank = 1 + #{tokens scored above the true one}



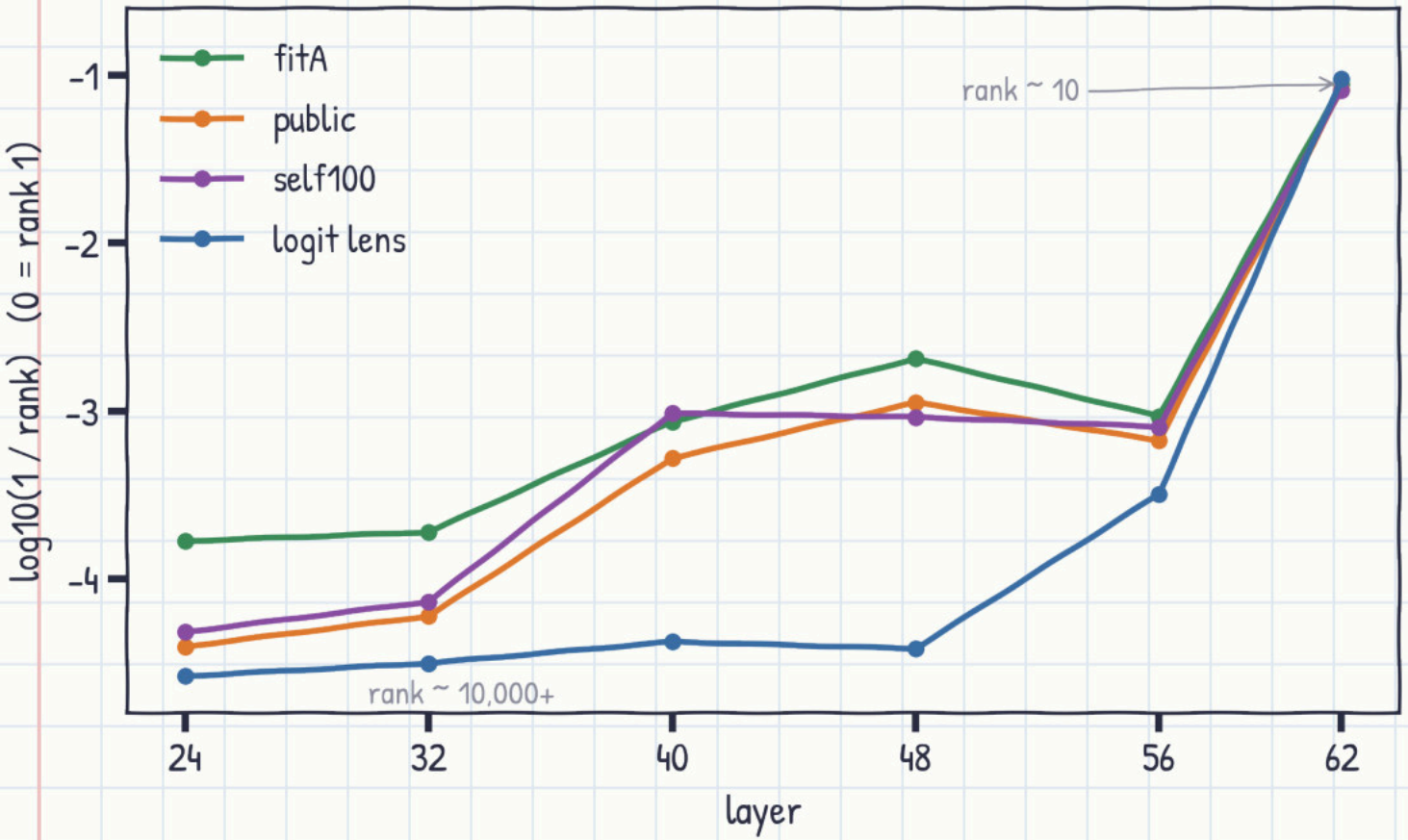


…titled | Affect | dynamics, | … | (2007)? | Authors | : | Kar | in | G | . | Co | if | man

Ranks of "Co" (author 1, token 5), true title: 351_stage10/raw/logit/L{24…62}/shard_011.npz row 25029; out = raw/final/L63

Names become readable late

all 9,108 papers · true title (Stage 10 cohort, §6.1)



Logit lens: almost nothing until **L48**

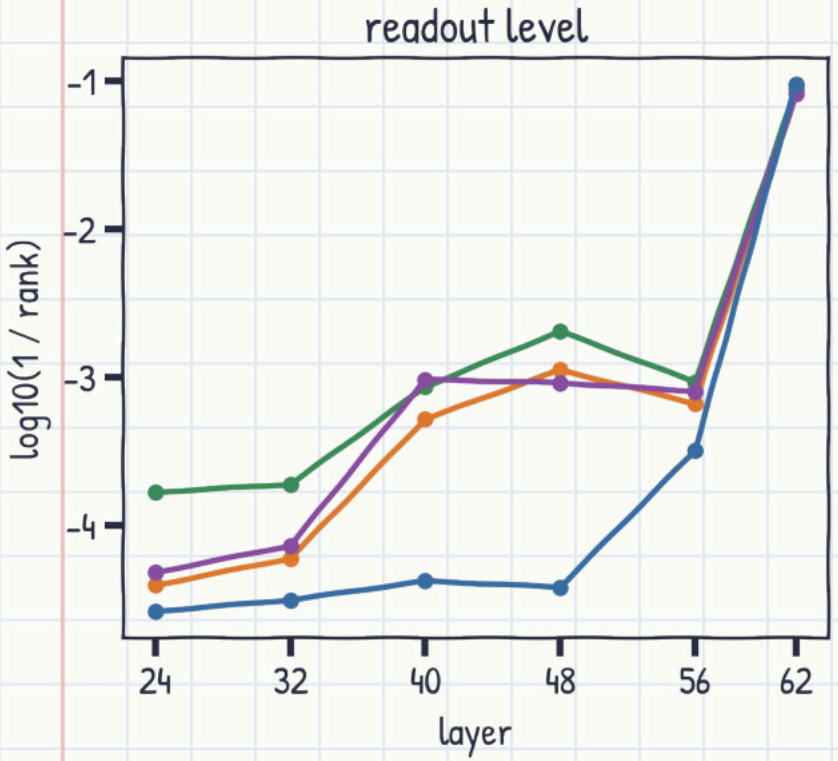
Jacobian lenses read more from **L40**

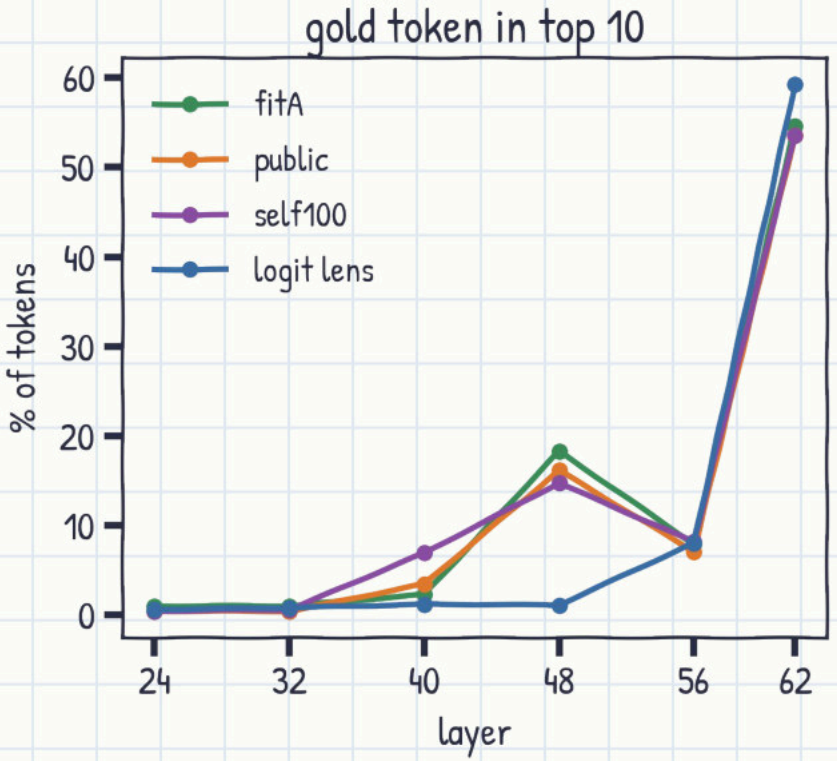
Everything jumps at **L62** (rank ~10) (Stage 10, logit L62 -0.93, §6.1)

Stage 10

Which lens reads best?

all papers · true title





Middle layers:

fitA best; at L48 15–18% top-10 vs 1% for logit

(Stage 10 X3, 9,108 papers, §6.9)

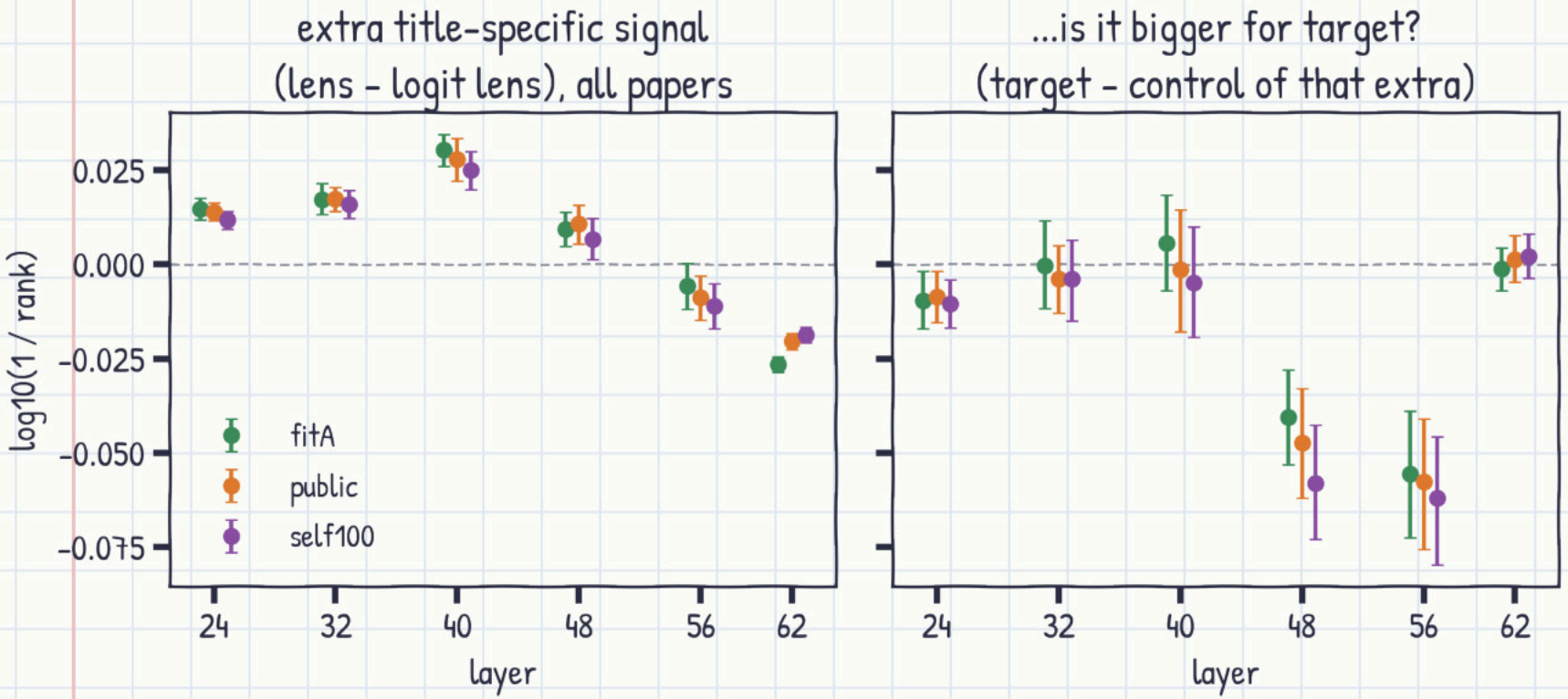
L62: logit lens best (59%)

(same, L62)

Stage 10

Do lenses reveal hidden paper knowledge?

paired differences, 95% CI



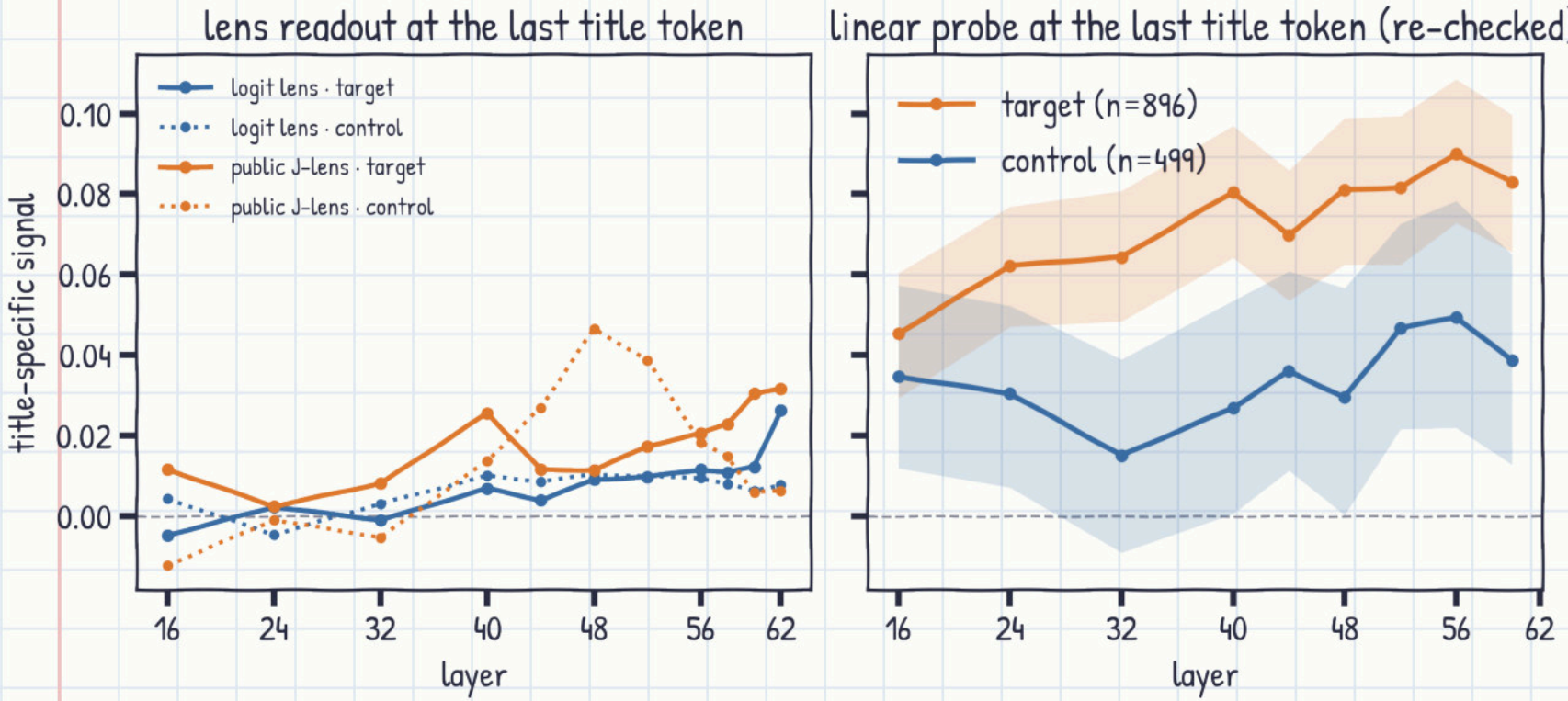
Extra signal: **+0.01 to +0.03** (Stage 10 X3, L24–L48, 9,108 papers, §6.9)

vs ~0.44 at L62 (same, L62)

Not bigger for target
(even negative at L48–56)

Read early: a probe finds it at the title

title-specific signal at the last title token · lenses: all 9,108 papers · probe: re-checked, title reuse removed: 1,863 papers (896 target, 499 control) (Stage 11 E4; Stage 13 E4b)



lenses: **~0.01–0.03** (Stage 11 E4, 4,621 + 2,220, §6.11)

probe: **0.05 -> 0.08** (Stage 13 E4b, 1,863 papers, first name token only)

target > control by **+0.03–0.08** (E4b, 896 vs 499, within bin)

re-checked: team folds, title reuse removed

scores the first name token only (mostly an initial); subtracts false_0 only

read early (title, from L24), **spoken late** (names, L57–61)

Probe scores only the first name token (mostly an initial); all-token, by-role version (addendum 9, done): surname first token L40 +0.071 [0.058, 0.085] (1,395 papers; e4b_alltok.md).

Stages 1–9 · 20–200 papers

What we saw

- Self-100 fits the final distribution better; author ranking not better
- J-Lens − Vanilla ≈0 (7.4: +0.002; Stage 8: −1.4 pp)(50 / 200 papers)
- Stage 9: only Fit A fitted, stability not tested

Stages 10–11 · 9,108 papers

What scale showed

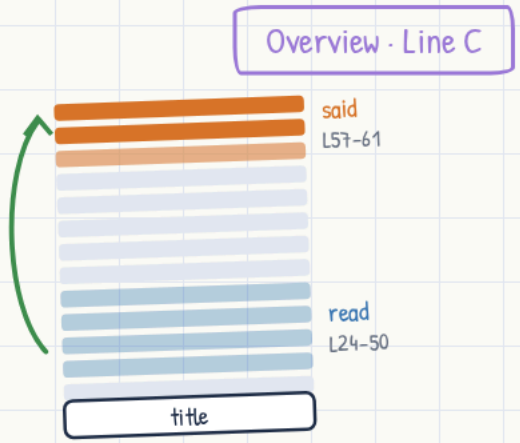
- Jacobian lenses: top-10 rate 15–18% at L48 vs 1% for logit lens(9,108 papers; X3_means.csv)
- Extra title-specific signal only +0.01–0.03(L24–L48; X3_means.csv), not target-specific
- At L62 the plain logit lens is best
- Linear probe at the title token: +0.05 → +0.08 (re-checked) (E4b, scores only the first name token; e4b_trainstd_compare.md), paper-specific

Line B: the information is there, but not in a form the lenses read

Both phases agree: no lens reveals hidden paper knowledge in middle layers. Scale adds the positive side: a trained probe can decode it at the title, so the limit is the readout, not the absence of information.

Line C

When does it appear, and when is it used?



small samples asked

Which layer peaks? (L56 kept standing out)

→

the full cohort answered

Readable late (L57–61) but read from the title early (L24–50)